

Context-Aware Linguistic Steganography: A Generative Workflow using Large Language Models

NASOUH ALOLABI, Higher Institute for Applied Sciences and Technology, Syria

RIAD SONBOL, Higher Institute for Applied Sciences and Technology, Syria

This paper proposes a context-aware **black-box** linguistic steganography workflow for threaded discussions that carries the payload through a *selection channel* rather than token-level manipulation. The sender reconstructs a shared context corpus, selects both a reply target and a semantic angle, and uses a stochastic LLM to realize the choice as an ordinary visible reply. The receiver rebuilds the same context, recovers the chosen selections, and decodes the payload from the published comment. The method is designed to remain natural in conversation, avoid visible control markers, and support recovery using only shared assumptions and the agreed temporal window. We evaluate the approach using naturalness and distributional divergence metrics on successful runs, and report generation and decode failures separately.

Additional Key Words and Phrases: Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Semantic Embedding, Agentic Workflows, Imperceptibility

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1 INTRODUCTION

Linguistic steganography focuses on the covert embedding of secret information within ostensibly innocuous textual carriers. Traditional methodologies, primarily predicated on lexical substitution and synonym manipulation [1], often struggle with the inherent low redundancy of natural language. Consequently, even minor statistical deviations can provide sufficient signal for modern automated detection systems [2–4]. The emergence of Large Language Models (LLMs) has catalyzed a paradigm shift, transitioning the field from text modification to de novo text generation [5–7]. By sampling directly from a model’s learned probability distribution, these generative approaches theoretically achieve high levels of statistical imperceptibility.

However, existing LLM-based steganographic systems remain constrained by a fundamental trade-off between statistical security and perceptual naturalness [5–7]. While a model may output grammatically and statistically fluent sentences, these often lack the deep semantic coherence required for specific communicative contexts, such as threaded social media discussions [8–10]. A comment that is technically flawless yet pragmatically irrelevant to the ongoing discourse serves as a conspicuous anomaly under human or semantic-level scrutiny.

To address these challenges, we introduce a context-aware steganographic framework that elevates the embedding domain from the lexical level (individual tokens) to the semantic level (conversational decisions). Instead of manipulating token distributions, our scheme carries the payload through a *selection channel*: bits map to (i) the index of the reply target within a deterministically flattened comment tree, and (ii) the index of a context-conditioned *Angle* within a shared angle set that both parties reconstruct deterministically from the public thread and reproducible research. A language model then generates an ordinary visible reply that serves as a *decodable witness* of the selected Angle; we do not rely on invisible characters or control Unicode. A sender-side verification loop closes the system by simulating

Authors’ addresses: Nasouh Alolabi, Higher Institute for Applied Sciences and Technology, Damascus, Syria; Riad Sonbol, Higher Institute for Applied Sciences and Technology, Damascus, Syria.

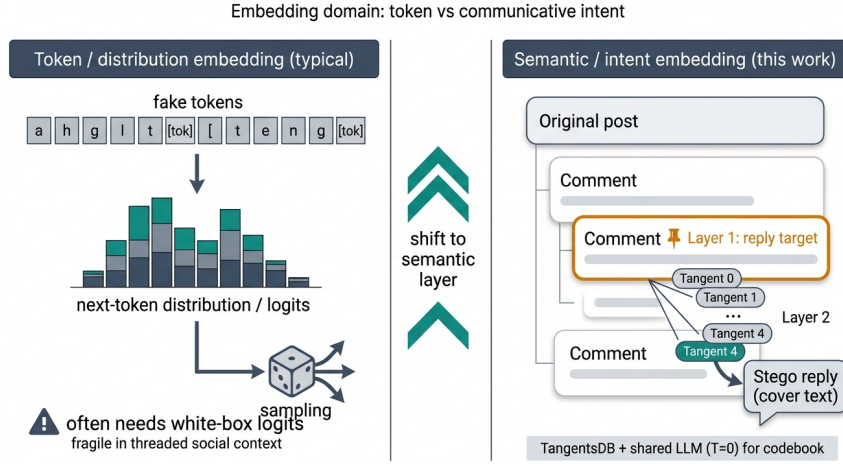


Fig. 1. Conceptual contrast between token-level embedding and our two-layer semantic embedding over a threaded discussion.

receiver decoding before publication. Figure 1 contrasts token-level embedding with our two-layer selection-channel embedding over a threaded discussion.

The primary contributions of this paper are:

- A two-layer, selection-channel embedding architecture for threaded discussions that encodes bits into reply targeting and Angle selection, with visible text serving as a decodable witness rather than a hidden carrier.
- A deterministic reconstruction pipeline (thread + reproducible research) that enables sender/receiver parity without white-box model access or side-channel metadata exchange.
- A closed-loop verification mechanism that simulates decoding pre-publication and retries synthesis to improve recoverability under stochastic generation.

2 BACKGROUND

Among the classical concerns of information security—protecting content through encryption, preserving user privacy, and concealing the existence of communication itself—this work is concerned with the last. **Steganography** differs from encryption in a fundamental way: rather than making a message unreadable, it makes the message invisible by embedding it inside an ordinary carrier, such as an image or a piece of text, so that an observer has no reason to suspect anything is hidden. **Imperceptibility** is therefore the primary objective, not confidentiality per se.

Text is a particularly challenging carrier due to its low redundancy and strict semantic constraints. The classical “Prisoners’ Problem” [11] illustrates the goal: two parties, Alice and Bob, must exchange hidden information without alerting a watchful adversary.

Textual steganography methods are typically divided into **format-based** approaches, which exploit layout or structural features, and **linguistic-based** approaches, which modify linguistic form. Within the latter, early techniques such as **synonym substitution** embed bits by altering lexical choices, but suffer from low capacity and high detectability.

More formally, generative linguistic steganography operates through an **embedding function** f_{emb} that establishes a cryptographic and linguistic mapping to hide secret messages within generated text. The algorithm takes as inputs a language model $\mathcal{L}$ and a secret message m (represented as a uniformly distributed bitstream). At each generation

step, rather than sampling freely from the language model’s conditional distribution p_{LM} , the embedding function f_{emb} **restricts or modifies** the token selection process according to the message bits being embedded. This specialized process is known as **steganographic sampling**, producing a steganographic text (stegotext) y . The security of the system depends on minimizing the divergence between the original language model distribution p_{LM} and the modified distribution q induced by f_{emb} [12].

Traditional linguistic approaches offer limited embedding capacity and often leave statistical artifacts. Advances in deep learning and **Large Language Models (LLMs)** now enable generative methods that achieve higher text quality and more secure embedding. Evaluating such systems requires several dimensions of imperceptibility: **perceptual** (human naturalness), **statistical** (distributional similarity to natural text), and **cognitive** (semantic and contextual fidelity) [13].

LLM use in steganography also introduces new sources of detectability. **Hallucination**—where a model generates content that is factually unsupported, semantically inconsistent, or poorly grounded in the preceding context—is one such source. In ordinary text generation, hallucination is mainly a reliability concern. In steganographic generation, it becomes a security concern as well: even a fluent sentence can expose a hidden channel if it contains an unsupported claim, an abrupt topic shift, or a contradiction with the surrounding context. Hallucination therefore threatens **cognitive imperceptibility**, since generated text must remain contextually plausible and communicatively coherent, not merely surface-fluent.

Distinct from hallucination, the **PSIC effect** (Perceptual-Statistical Imperceptibility Conflict) [2] captures a different and arguably more fundamental tension in generative linguistic steganography. Text that is optimized for human fluency is not necessarily statistically indistinguishable from natural human text. For example, selecting only high-probability tokens can produce sentences with low perplexity and strong local readability, but it may also make the generated distribution overly smooth, regular, or concentrated compared with real human writing. Such text can appear natural to readers while remaining detectable to statistical steganalyzers [2].

The PSIC effect therefore shows that **perceptual imperceptibility** and **statistical imperceptibility** are related but not identical goals. Human-written text often contains variation, irregularity, and locally unexpected word choices. By contrast, overly fluent model-generated text may lack this natural variance. Increasing the embedding rate can sometimes force the sampler to select from a broader range of tokens, which may better approximate the diversity of human text; however, this same process can also reduce readability or semantic coherence if the token choices become poorly aligned with the context. Thus, higher capacity does not automatically imply better security. Secure generative steganography requires balancing embedding rate, fluency, distributional similarity, and contextual consistency.

2.1 Large Language Models as Text Generators

Large Language Models (LLMs) are autoregressive, generative systems based on the Transformer architecture [14] that approximate high-dimensional distributions over natural-language sequences [3, 15]. Given a prefix, an LLM emits a probability vector over the vocabulary; the next token is sampled from this vector and appended to the prefix, and the process repeats until a stopping criterion is met. During pre-training, billions of parameters are tuned on large text corpora so that the model’s predictive distribution approximates patterns in the training data [16]. As a consequence, modern LLMs can produce text with high fluency, coherence, and stylistic control [17]. The learned representations capture stylistic and semantic regularities that generalize across domains, enabling applications requiring nuanced linguistic mimicry [18].

2.2 Large Language Models in Generative Linguistic Steganography

Because text produced by modern LLMs can closely resemble ordinary human-authored text, these models provide effective engines for generative text steganography. Rather than editing an existing cover, a growing line of work treats the model itself as a steganographic sampler: the hidden message steers generation, so the stego text is drawn from a realistic communication distribution from the start. The public availability of high-quality models, together with the efficiency of sampling-based embedding, has made this route more practical than earlier designs.

LLMs like **GPT-2** [15], **LLaMA** [19], and **Baichuan2** [20] are commonly used as basic generative models for steganography. Most such schemes couple the language model with a steganographic code that ties message bits to token choices: at each generation step, the bits to be hidden determine which candidate token is emitted from the model’s next-token distribution. However, traditional “white-box” methods necessitate sharing the exact language model and training vocabulary, which limits fluency, logic, and diversity compared to texts generated by modern black-box LLM APIs. These methods inevitably alter sampling probability distributions when embedding messages, posing security risks [7]. **Black-box approaches** avoid these limitations by leveraging state-of-the-art hosted models to deliver better text quality, faster generation speeds, and minimal local resource requirements. However, this paradigm trades fine-grained control over internal sampling for these practical advantages. Conversely, **white-box access** enables precise parameter control and stronger security guarantees, but demands substantial computational resources, engineering effort, and higher latency—raising deployment barriers. This trade-off between access paradigms is central to evaluating the robustness and applicability of modern linguistic steganography, and it directly motivates the black-box design adopted in Section 4.

New approaches, such as **LLM-Stega** [7], explore black-box generative text steganography using the user interfaces (UIs) of LLMs, which removes any need to observe internal sampling distributions. Secret bits are encrypted and mapped onto a prearranged keyword set that steers generation, and candidate outputs from which the message cannot be recovered are discarded via reject sampling, preserving both extraction accuracy and semantic quality [7].

Another framework, **Co-Stega**, leverages LLMs to address the challenge of low capacity in social media. It expands the space available for hiding messages through context retrieval and uses purpose-built prompts to raise the entropy of the generated replies, which in turn enlarges the embedding capacity while keeping the output fluent and topically relevant [8].

Zero-shot linguistic steganography instead relies on in-context learning: covert text samples are supplied in the prompt, and the stego text is produced within a question–answer (QA) exchange, so the output reads as a natural reply rather than as free-standing generated text [6].

The increasing popularity of deep generative models has made it feasible for provably secure steganography to be applied in real-world scenarios, as they fulfill requirements for perfect samplers and explicit data distributions [3, 21, 22].

LLM-based steganographic methods are typically evaluated on two primary axes: imperceptibility (perceptual, statistical, and cognitive measures) and embedding capacity (e.g., bits per token or bits per word). Imperceptibility evaluations may include automatic metrics (PPL, Distinct-n, KL/JSD) as well as human judgements; embedding capacity is usually reported as bits/token or overall embedding rate. Section 3 surveys how the literature applies, evaluates, and constrains these methods in more detail.

Table 1. Comparison of the literature synthesis in this section with prior surveys of text/linguistic steganography.

Survey	Period covered	Methodology	Carrier scope	LLM-era coverage	Contextual-compatibility focus
Majeed et al. [23] (2021)	Pre-transformer	Narrative	Text-only (synonym, spacing, coding)	None	No
Setiadi et al. [24] (2025)	2021–2025	Narrative	Multi-carrier (image, linguistic, 3D mesh)	Yes	No
Wang et al. [4] (2023)	Deep-learning era	Narrative	Detection-side (linguistic steganalysis)	Partial	No
This synthesis	2018–2025	Systematic	Text-only (LLM-based)	Yes	Yes

3 RELATED WORK

Majeed et al. (2021) [23] surveyed pre-LLM text steganography techniques, predating the current transformer era, emphasizing synonym replacement, spacing manipulation, and coding schemes such as Huffman coding. The methods of that generation often relied on context-free grammars or Markov chains and tended to produce syntactically correct but semantically incoherent cover text. Setiadi et al. (2025) [24] examine more recent AI-powered data hiding across image, linguistic, and 3D mesh carriers (2021–2025), observing that LLMs have “revitalized” linguistic steganography through methods built on GPT-2 [25], GPT-3 [16], LLaMA2 [26], and Baichuan2 [27]. On the detection side, Wang et al. [4] review deep-learning linguistic steganalysis and document how detection techniques have evolved alongside generative hiding methods.

These works provide valuable coverage of text steganography, but they are best described as narrative surveys rather than exhaustive systematic literature reviews, and none of them foregrounds *contextual compatibility*—the degree to which generated cover text fits its surrounding discourse, task, and conversational setting—as a central evaluative criterion for LLM-based methods. Crucially, the field has evolved from “statistical vector embedding” (Word2Vec, GloVe) to “language-model vector embedding” that exploits BERT-scale transformers and higher-dimensional semantic spaces. This creates a methodological gap: no systematic review comprehensively maps how large-scale transformers have redefined text steganography along this dimension. Modern advances extend beyond naive generation to sophisticated Controllable Text Generation (CTG) frameworks [28]. These employ Variational Autoencoders (VAEs) to model latent features and diffusion models to inject randomness, mitigating spurious associations between secrets and control conditions. Contemporary approaches leverage prompt learning and prefix tuning, enabling efficient model customization without costly full fine-tuning.

Defensive strategies must evolve accordingly. Traditional steganalysis, premised on hand-crafted statistical features, falters against generative steganography’s high statistical concealment. Current research must confront “stegomalware”—attacks that conceal command-and-control communications within innocuous digital media [29].

To close this gap, the remainder of this section synthesizes findings from a systematic review of 31 primary studies on LLM-based linguistic steganography published between 2018 and 2025, organized around the state of the published literature, its application domains, its evaluation practices, its patterns of external-knowledge integration, and its recurring limitations and trade-offs. Table 1 summarizes how this synthesis differs from the three prior surveys discussed above.

Table 2. Representative model families across the reviewed studies. Categories are non-exclusive, so totals exceed 100%.

Model Type (Studies)	Models and Representative Works
Open-weight Models (25/31, 80.6%)	GPT-2 [3, 9, 21, 36], LLaMA/LLaMA2 [6, 8, 22, 37], BERT [1, 10, 13, 28, 38–42], OPT [43], BART [1, 5], Qwen [33, 44]
Proprietary Models (7/31, 22.6%)	GPT-3.5/4, ChatGPT [7, 30–32, 45, 46]
Custom or Task-Specific Architectures (5/31, 16.1%)	From-scratch or task-specific models [2, 35, 39, 40, 43]

3.1 State of the Published Literature

The 31 studies span 2020 through 2025, with 10 publications falling in 2020–2023 and 21 in 2024–2025 alone—a sharp acceleration that coincides with the wider availability of large-scale models. Widely used model families—GPT, GPT-2, and LLaMA [15, 16, 19]—appear across both periods, though proprietary systems grow more common in the later studies as black-box access becomes a viable design choice. Across the reviewed corpus, open-weight and research-accessible models appear more frequently than closed commercial systems, reflecting the practical importance of reproducibility and controllable experimentation.

The field has also evolved from earlier white-box token-sampling schemes toward a broader design space that includes hybrid and black-box methods. The reviewed studies now span generative schemes that write stego text from scratch [2, 3, 12, 21], rewriting systems that paraphrase existing covers [5], black-box pipelines that treat the model as an opaque API [7, 30–32], zero-shot protocols driven primarily by prompting [6], context-aware frameworks that retrieve or constrain external information to increase semantic plausibility or payload [8, 9, 33], and methods that emphasize stronger security, robustness, or token-level quality control [3, 21, 34, 35].

Table 2 shows how the reviewed studies distribute across model families. This table is best read as a synthesis of methodological tendencies rather than a strict one-study-one-category classification: many papers combine multiple model families, so the counts overlap and the percentages do not sum to 100%. Open-weight models account for the largest share because they support reproducible experiments and direct control over sampling or decoding, with GPT-2 and LLaMA-family systems appearing especially often. BERT-based methods also recur frequently, but usually as masked-language or scoring components rather than as the sole generation backbone. Proprietary systems such as GPT-3.5/4 are mainly used in black-box or prompt-driven settings, while custom or task-specific architectures appear less often but remain important for studies that prioritize provable security, reversible encoding, or tightly controlled payload design.

3.2 Application Domains

The reviewed studies show that **covert communication** remains the dominant application of LLM-based linguistic steganography. In this setting, secret information is embedded into seemingly benign text so that the message can pass through ordinary communication channels with minimal suspicion. LLM-based methods improve this use case by producing more fluent and contextually plausible outputs than earlier rule-based or local-substitution systems [2, 3, 7].

Within this application space, the reviewed papers explore several operational variants. Some methods generate stego text from scratch using a shared model or sampling strategy [3, 21]. Others rely on paraphrasing or rewriting so that a pre-existing cover text can be transformed while preserving its apparent meaning [5]. Recent black-box and prompt-based approaches such as **LLM-Stega**, **DeepStego**, and multi-criteria linguistic optimization show that covert communication can also be implemented through commercial or hosted interfaces without direct access to token probabilities, using prompt design, keyword sets, stylistic features, and reject-sampling style verification instead

Table 3. Application categories across the reviewed studies. Categories are non-exclusive.

Primary Application	Variant / Category	Representative Studies	Total Count
Covert Communication	Generation-based (from scratch)	[2, 3, 21, 22, 28, 34–36, 39, 41, 42]	11
	Paraphrasing, Rewriting, or Editing	[5, 38]	2
	Black-box and Prompt-based	[6, 7, 30–32, 37]	6
Context-Aware Systems	Domain, Emotion, or Social Adaptation	[8–10, 13, 33, 44]	6
Watermarking	Content Attribution (Boundary Cases)	[1, 40, 43, 45–47]	6

[7, 31, 32]. Context-aware systems such as **Co-Stega**, **Hi-Stega**, and emotionally controllable steganography further adapt the generated text to specific domains, topics, entities, emotional tones, or social settings in order to improve plausibility and increase effective payload [8, 9, 33]. This last variant—adapting generation to the surrounding domain and discourse rather than relying on generic fluency—is the closest precedent in the literature to the context-aware, thread-level embedding developed in Section 4.

Although adjacent topics such as watermarking, authorship attribution, or prompt-hiding attacks are relevant to the broader field of information hiding, they remain analytically distinct from covert payload transmission. The 2025 watermarking studies are therefore treated as boundary cases: useful for understanding robustness, semantic preservation, and detection methods, but not interchangeable with steganographic communication systems [47].

A comprehensive breakdown of how the reviewed studies are distributed across these primary application areas and variants is summarized in Table 3.

3.3 Evaluation Practices in the Literature

Performance evaluation for LLM-based steganography relies on three main categories of metrics, but reporting standards vary widely across studies. This variation limits direct comparison and motivates more consistent evaluation protocols; it also directly informs the choice of metrics used for this paper’s own evaluation in Section 6.

3.3.1 Perplexity. Perplexity is an imperceptibility metric for fluency, where lower values generally indicate more natural text [48]. However, its usefulness for language model evaluation is limited. It mainly reflects internal consistency rather than factuality or reasoning, so fluent but implausible statements can still receive low perplexity scores [49]. It is also sensitive to text length, with short sequences often receiving inflated scores even when they are highly fluent [50]. In addition, perplexity is not directly comparable across models unless tokenization and vocabulary are aligned, since these design choices affect the score itself [51]. Fang et al. [50] further argue that token-averaged likelihood can under-represent performance on important low-probability tokens, while Wang et al. [49] note that repetition can sometimes lower perplexity in misleading ways. Taken together, these issues suggest that perplexity should be used cautiously and alongside other measures when assessing steganographic text quality.

3.3.2 Statistical Divergence Metrics. Kullback-Leibler Divergence (KLD) [52] and Jensen-Shannon Divergence (JSD) are information-theoretic metrics used to evaluate steganographic security. KLD quantifies information loss by measuring the relative entropy between cover and stego distributions, serving as the theoretical standard for security modeling despite being asymmetric and failing as a strict distance measure. JSD improves upon this as a symmetric, bounded variant that measures how far each distribution lies from their average, providing a more stable basis for formulating

statistical imperceptibility bounds—particularly when language models approximate human text distributions. Together, these two metrics attempt to capture how closely steganographic outputs mimic legitimate communication channels.

However, real-world application reveals reliability limits, most notably the PSIC effect [2]. KLD and JSD scores can diverge from human judgment as statistical optimization progresses: methods with favorable divergence scores may still produce low-quality text that appears suspicious to human observers. This discrepancy is also dataset dependent—the same method can yield different divergence values on different corpora at equivalent embedding rates (e.g., KLDs of 19.507 on IMDB versus 8.295 on Twitter for VAE-Stega [2]), making cross-paper comparison unreliable. In addition, studies use different formulas, feature spaces, and measurement scales, including latent BERT features versus direct word distributions. Meteor [3], for example, is associated with reported KLD values ranging from 0.045 in one setting to 7.491–11.845 in others. Consequently, these metrics capture distributional distance but not perceptual naturalness, so they should be paired with semantic and human-centered measures.

3.3.3 *Capacity Metrics.* Capacity is judged by four metrics:

- **Bits per Token (BPT):**

$$\text{BPT} = \frac{\text{Total Secret Bits}}{\text{Total Tokens}} \quad (1)$$

- **Bits per Word (BPW):**

$$\text{BPW} = \frac{\text{Total Secret Bits}}{\text{Total Words}} \quad (2)$$

- **Embedding Rate (ER) [53]:** Average density of hidden information per textual unit

$$\text{ER} = \frac{1}{N} \sum_{i=1}^N \text{bits}_i \quad (3)$$

where N is the number of textual units (words, tokens, or sentences) and bits_i is the number of bits embedded in the i -th unit. In some fixed-payload schemes, the total number of embedded bits is predetermined for each stego text and does not depend on its length; for example, LLM-Stega can assign a constant payload to every generated stego sentence, so the embedding rate can also be expressed as $\text{ER}_{\text{fixed}} = B_0$, where B_0 is the fixed number of bits embedded in each stego text.

- **Utilization Rate (UR):**

$$H = - \sum_{x \in \mathcal{X}} P(x) \log_2 P(x) \quad (4)$$

$$\text{UR} = \left(\frac{\text{Actual Bits Embedded}}{H} \right) \times 100\% \quad (5)$$

These quantities describe how densely a secret is embedded, but they are affected by systematic biases that limit cross-system comparison, summarized in Table 4.

Tokenization differences make “1 BPT” from one paper difficult to compare with “1 BPT” from another when the studies use different tokenizers. The PSIC effect shows that higher density can hurt human fluency yet help statistical evasion. Model frequency preferences shrink the real alphabet to high-probability tokens, so naive entropy limits overstate usable space. Ambiguous reporting—practical vs. effective payload, or bit length vs. stego length—can make capacity claims appear stronger than they are. Finally, BPW/BPT ignore semantics, so high values can be reported even when the resulting text is suspicious or incoherent. Recent works reveal additional distortions: loop-count overhead, dictionary-size caps, baseline-dependent “improvements,” watermarking goals that invert the desired signal, conversational filler that dilutes BPW, and nonlinear ER curves that make any single threshold difficult to interpret. The

Table 4. Five primary bias categories affecting capacity metrics in steganographic evaluation.

Bias Category	Core Problem	Critical Implication
Tokenization Inconsistencies	Metrics depend entirely on specific tokenizers (e.g., GPT-2 BPE vs. word-level)	Direct comparisons across papers become meaningless when tokenization strategies differ
The PSIC Effect	Conflicts between imperceptibility and statistical security are ignored	High capacity may degrade human fluency while paradoxically improving detection resistance
Model Training Bias	Utilization Rate calculations assume uniform token availability	Actual hiding space is smaller than theoretical entropy due to model frequency preferences
Reporting Ambiguities	No standard definition of “capacity” across systems	Practical payload vs. effective payload distinctions create misleading efficiency claims
Context Blindness	Density metrics treat text as neutral bit containers	Semantic incoherence constitutes a security failure that BPW/BPT fails to penalize

Table 5. Representative capacity calculations reported by selected methods. Full generated passages are omitted because they do not support direct cross-study comparison.

Method	Reported Capacity	Encoding Basis	Comparability Note
Meteor	1280 bits	160-byte payload $\times$ 8 bits	Payload total; depends on the selected message length.
LLM-Stega	64 bits per sentence	Fixed semantic-slot mapping	Sentence-level capacity; not directly comparable with BPW or BPT.
GCStego	16 bits per word	Fixed word-level mapping	Mapping-specific rate under the study’s candidate construction.
ParaLS	3 bits	Binary lexical watermark	Example payload rather than a normalized embedding rate.
Zero-shot	24 bits per four Base64 characters	Encoded secret representation	Capacity depends on the representation and prompt protocol.
Discop	968 bits over 200 tokens	4.84 bits per token at $p = 0.95$	Token-level rate tied to the model, tokenizer, and truncation setting.

added 2025 studies reinforce this point: DeepStego emphasizes high capacity and recovery accuracy, and multi-criteria optimization reports reliable stylistic decoding [31, 32], but these results also make cross-paper comparison harder because the reported gains depend on different baselines, prompts, datasets, and decoding assumptions.

To contextualize theoretical capacity metrics, Table 5 summarizes representative embedding capacities across the reviewed methods. Capacity calculations vary significantly based on the encoding mechanism. **Fixed-length mapping** techniques, such as **LLM-Stega** and **GCStego**, derive capacity from rigid keyword or token mappings—64 bits per sentence and 16 bits per word, respectively. **ParaLS** operates similarly but on a smaller scale, encoding a simple 3-bit binary watermark. In contrast, **payload-dependent** methods scale with the cover text or secret message length: **Meteor’s** capacity of 1280 bits corresponds to an embedded payload of 160 bytes, **Zero-shot** capacity is directly tied to the Base64 representation of the secret bitstream, and **Discop** uses a statistical approach that, with a truncation parameter of $p = 0.95$, achieves an entropy rate of 4.84 bits per token, yielding 968 bits over a 200-token sequence.

3.3.4 Security Metrics. The sources evaluate security and reliability through the following integrated metrics:

- **Detection Accuracy (Acc) / Error Rate (PE):** Measures a classifier’s ability to distinguish between cover and stego text. An accuracy of 50% (or PE of 50%) is the gold standard, indicating the stego text is statistically identical to normal text.
- **F1 Score:** The harmonic mean of precision and recall, used to verify detection reliability, especially when datasets are balanced.
- **Robustness (Attack Resistance):** Evaluated via the Mean Impact of Attack (mIOA) or bit accuracy after removal attempts such as denoising auto-encoding, word substitution, or sentence deletion.
- **False Positive Rate (FPR):** Measures how often human or normal model-generated texts are incorrectly flagged as containing secret messages.

Table 6. Frequency of evaluation metric categories in the reviewed studies. Categories are non-exclusive.

Dimension	Metric Category	Studies	Main Comparability Constraint
Imperceptibility	Perplexity	13	Model, tokenizer, corpus, and sequence length
Imperceptibility	Statistical divergence (KLD/JSD)	10	Estimated distributions and sampling settings
Imperceptibility	Semantic similarity	13	Reference construction and metric choice
Imperceptibility	Human evaluation	1	Protocol, annotator population, and sample size
Security	Detection accuracy or F1	21	Detector, class balance, and train/test data
Capacity	BPW or BPT	27	Tokenization and payload accounting
Robustness	Recovery or attack resilience	5	Attack type, strength, and evaluation protocol
Efficiency	Generation or extraction time	5	Hardware, model size, and implementation

Recent robustness-oriented work broadens this evaluation set. Position-agnostic generation tests perturbation tolerance across multiple attack types, active-attack repair methods evaluate message integrity after tampering, and contrastive semantic watermarking reports F1 under rewriting and substitution attacks [34, 35, 47]. This shift suggests that future evaluations should report not only clean-channel capacity and fluency, but also degradation curves under realistic editing.

3.3.5 Evaluation Challenges and Gaps. The broad range of evaluation metrics across the reviewed studies is summarized in Table 6, which categorizes metrics by type and quantifies their usage frequency. The numbers in that table reveal several persistent gaps. Shared benchmarks are rare: only around 20% of studies use common datasets, which makes cross-paper comparison difficult even when the same metric name appears in both papers. Human evaluation was reported in just one of the 31 reviewed studies—a surprisingly low share given how central perceptual plausibility is to the field’s core claims. Robustness testing is similarly sparse, absent in roughly 60% of studies. Many papers also evaluate along only one or two metric dimensions, leaving imperceptibility, security, and capacity to be inferred from partial evidence rather than directly measured together. Taken together, these gaps mean the strongest conclusions available from the reviewed corpus are qualitative rather than quantitative—a limitation this paper addresses directly by reporting naturalness, distributional divergence, and operational reliability jointly for the same runs (Section 6).

3.4 Integration of External Knowledge Sources

External knowledge integration is a recurring design pattern in the reviewed literature. Across the surveyed studies, external information serves three main purposes: improving contextual plausibility, maintaining topic consistency, and—in some cases—expanding the effective space available for embedding. Table 7 summarizes the main knowledge-integration patterns and their roles.

Rather than relying only on the internal prior of the language model, several recent systems inject external guidance during generation. Retrieved posts, headlines, or comments can provide a realistic discourse context for the cover text [9]. Knowledge-graph triples can constrain the content that should appear in the output, improving semantic consistency over longer spans [37]. Named entities and sentiment labels add a finer-grained control layer when the goal is emotional or conversational consistency [33]. Frequency-based external statistics can also be used to steer generation toward distributions that better match the target channel [36].

Table 7. Representative patterns of external knowledge integration.

Knowledge Type	Typical Role	Primary Benefit	Examples
Semantic Resources	Topic, entity, and emotion guidance	Better semantic control	Co-Stega [8], knowledge-graph guided methods [37, 39], emotionally controllable steganography [33]
Domain Corpora	Domain adaptation	Better stylistic fit	FREmax [36], specialized datasets
Prompt Engineering	In-context control	Better task alignment	Zero-shot [6], DeepStego [31], multi-criteria optimization [32]
Context Retrieval	Evidence or example retrieval	Higher contextual plausibility	Co-Stega [8], retrieval-augmented pipelines [9]

Another recurring pattern is the use of domain-specific corpora. Fine-tuning or conditioning on domain data helps align the generated text with the vocabulary, stylistic conventions, and topical expectations of the intended communication channel. This pattern appears in systems such as VAE-Stega [2], Meteor [3], Hi-Stega [9], FREmax [36], Rewriting-Stego [5], and Summarization-Stego [10]. When additional training is not possible, black-box methods move this domain adaptation into the prompt, style specification, or retrieval stage. Zero-shot approaches provide examples directly in the context window, LLM-Stega frames the generation request with topic-constrained prompts, DeepStego uses prompts and synonym guidance, multi-criteria optimization encodes through stylistic controls, and Co-Stega retrieves recent posts to anchor the generated reply in a plausible local context [6–8, 31, 32]—the same retrieve-then-anchor strategy that underlies the deterministic research stage described in Section 4.3.

Table 8 provides a more detailed breakdown of how external knowledge sources are integrated across the reviewed studies.

These five patterns differ primarily in how directly they constrain the generation process. Structured knowledge approaches [37, 39] use external knowledge graphs or graph attention networks to enforce long-range semantic consistency—a stronger form of control than prompting alone, but one that requires the knowledge resource to exist and be aligned with the target domain. Information retrieval frameworks [8, 9] pull contextually relevant material from external sources at runtime, grounding the cover text in real discourse without requiring model retraining. Linguistic resource methods [1] rely on synonym databases and paraphrase corpora to constrain lexical choice while preserving meaning—a lighter form of intervention that nonetheless depends on the quality and coverage of the external resource. External model feedback is the most common pattern, appearing in seven of the 31 reviewed studies: auxiliary models such as BERT serve as discriminators or soft scorers to assess naturalness or security during generation [41, 43]. Finally, shared side information methods [3, 6] use public models or auxiliary history to synchronize probability distributions between sender and receiver without explicitly passing knowledge through the channel, trading interpretability for coordination efficiency—the same trade-off that motivates the shared, deterministic search pipeline in Section 4.3.

3.5 Limitations and Trade-offs

Across the surveyed studies, the main constraints recur around imperceptibility, effective payload, semantic coherence, tokenization reliability, deployment assumptions, and attack robustness.

3.5.1 The PSIC Effect Revisited. Generative steganography faces an intrinsic tension: optimizing for perceptual fidelity degrades statistical security. Aggressive perplexity minimization produces unnaturally “sharp” distributions that deviate from high-variance human writing, increasing detection vulnerability. Higher embedding rates force selection of lower-probability tokens, degrading text quality but expanding the candidate pool to enhance anti-steganalysis resistance. At

Table 8. Comprehensive overview of external knowledge integration patterns in LLM-based steganography.

Approach	Variant / Category	Representative Studies	Count
Structured Knowledge Integration	Knowledge Graphs (KG) & Graph Attention Networks	Semantic controllable steganography via knowledge graph [37], joint linguistic steganography with BERT and graph attention networks [39]	2
	Information Retrieval (IR): Retrieval-Augmented Generation (RAG) & Corpus Search	Hi-Stega [9], Co-Stega [8], emotionally controllable text steganography [33]	3
Linguistic Resource Integration	Paraphrase DBs, Synonym DBs & Dependency Datasets	Paraphraser-based lexical substitution [1], DeepStego [31], post-hoc watermarking of black-box LLM text [46], group-chat imperceptible text steganography [44]	4
	External Statistical Analysis: Character/Token Frequency Distributions	FREmax [36]	1
External Model Feedback	Auxiliary Scorer, Critic & Discriminator Models	Rewriting-Stego [5], principled natural language watermarking [40], CPG-LS [41], controllable semantic steganography via summarization [10], ALiSa [42], DeepTextMark [43], contrastive semantic watermarking [47]	7
Shared Knowledge / Side Information	Shared Auxiliary Datasets & Public Contexts	Meteor [3], zero-shot generative linguistic steganography [6], reversible data hiding via masked language modeling [38], natural language steganography by ChatGPT [30]	4

elevated capacities, output distributions asymptotically approach natural language’s high-entropy chaos, camouflaging signals within legitimate statistical noise [2].

3.5.2 Attack Vulnerability and Security Concerns. Current techniques exhibit a substantial attack surface across multiple vectors, including word-level deletions and lexical substitutions that disrupt dependency parsing. Polishing and re-translation attacks progressively erase watermarks; at high intensities, these structural changes can reduce detection performance to an F1 score of 0.5, rendering the watermark undetectable. Furthermore, adversarial machine learning enables the extraction of secret model parameters from black-box systems, potentially compromising integrity entirely. Recent work responds by moving robustness into the embedding design itself: position-agnostic generation avoids dependence on fixed token locations, active-attack repair models attempt to recover damaged stegotext, and contrastive semantic watermarking tests resilience under rewriting and substitution [34, 35, 47]. To mitigate distributional risks, Variational Auto-Encoders are also employed to learn the statistical distribution of normal text, aiming to reduce the KL divergence between cover and stego distributions and resolve the conflict between perceptual and statistical imperceptibility [46].

3.5.3 Capacity and Contextual Constraints. Short, low-entropy contexts severely constrain embedding capacity. Social media posts and technical documents offer minimal redundancy for hiding [8], while semantic preservation requirements

compete directly with information embedding objectives. Brief texts further lack sufficient contextual support for effective steganographic operations [13]. Newer approaches attempt to recover some capacity through synonym guidance and stylistic feature control, but these gains remain tied to the selected prompt, cover text, or candidate-token space [31, 32].

Early generative methods also produced fluent but semantically arbitrary text, violating cognitive imperceptibility [13]. When generation is conditioned only on local token history, the output may drift away from the source topic or conversational logic, which is especially suspicious in low-entropy channels such as social media replies. This contextual drift worsens when methods must keep generating until an arbitrarily large payload is fully embedded: texts become unnaturally long, coherence decays over time, and practical payload efficiency drops even when reported bit capacity appears high [30, 37, 41, 45].

3.5.4 Segmentation and Tokenization Barriers. Subword tokenization introduces critical extraction ambiguities that are fundamental to white-box steganographic methods. Byte-pair encoding splits words unpredictably, creating multiple valid segmentations that corrupt message boundaries and cause decoding failures. These ambiguities impose hard capacity caps regardless of algorithmic sophistication—SparSamp suffers accuracy degradation from token ambiguity, while ShiMer cannot effectively boost entropy due to tokenization constraints [22].

This segmentation ambiguity persists even when sender and receiver employ identical LLMs and tokenizers. Because modern LLMs decompose words into subword units, and vocabularies frequently contain both complete words and their constituent components, any given text string admits multiple valid token representations. This multiplicity is compounded by the detokenization-retokenization gap: senders must convert tokens to plain text to conceal their transmission, while receivers must subsequently retokenize this text to recover hidden messages. Despite shared tokenization rules, receivers may select different token sequences than senders intended—a sender might encode a message using two separate tokens (“any” and “thing”), only for the receiver to retokenize the surface string as the single token “anything”. Because LLMs operate autoregressively, a single divergent token choice alters the probability distribution for all subsequent predictions, causing total decoding failure for the remainder of the message. To address this white-box-specific challenge, researchers propose SyncPool, a mechanism that aggregates ambiguous tokens into pools and employs synchronized random number generators to ensure both parties select identical tokens [22]. Because this work embeds payload through reply targeting and angle selection rather than through raw token identity (Section 4.6.1, Section 4.6.2), it sidesteps the detokenization-retokenization gap entirely: the sender and receiver only need to agree on which discrete angle or reply target was selected, not on an exact token sequence.

3.5.5 White-box versus Black-box Trade-offs. The architectural choice between white-box and black-box access paradigms involves fundamental trade-offs. Across the reviewed corpus, only a minority of studies operate primarily through black-box or prompt-driven access (Table 3), while the majority employ white-box or hybrid methods. White-box methods typically assume access to internal language-model components such as token probabilities, decoding states, vocabularies, or model parameters, and some approaches additionally require expensive fine-tuning or retraining procedures to align the model distribution with steganographic objectives. While these methods can achieve strong statistical imperceptibility through distribution-aware generation (e.g., VAE-Stega and arithmetic-coding-based methods), they suffer from reduced accessibility, high computational cost, and deployment complexity.

In contrast, black-box approaches operate through standard LLM APIs or user interfaces without requiring direct access to model internals or retraining, substantially improving practicality and usability. However, the distinction is primarily about *model access*, not the absence of probabilistic coordination, as several black-box approaches still

rely on shared probabilistic mechanisms, semantic distributions, or synchronized auxiliary structures between sender and receiver. Recent semantic black-box methods also prioritize robustness against paraphrasing, token perturbation, and noisy communication channels rather than purely token-level imperceptibility. Among the few peer-reviewed black-box methods currently available, LLM-Stega reports a capacity of 5.93 bpw with perplexity 165.76, illustrating that improved accessibility does not eliminate the PSIC trade-off between payload capacity, text quality, robustness, and security. Several recent systems further blur strict categorical boundaries by combining retrieval-, generation-, editing-, or substitution-based paradigms within hybrid steganographic frameworks.

3.5.6 Computational and Resource Constraints. Performance optimization conflicts directly with computational efficiency [3]. Superior results demand increased resources and memory, particularly for large models with external knowledge bases [21]. Real-time latency constraints limit optimization depth, while performance degradation emerges at operational scale. Variable length sampling, rejection sampling, active repair, and token-level quality control significantly increase encoding or decoding complexity, and LLM-based steganography requires 3x to 5x more time than prior methods [22]. The Meteor system exemplifies hardware constraints, running nearly 50× slower on mobile devices compared to GPU environments. LLM-specific deployment obstacles compound these costs: hallucinations and nonsensical continuations can corrupt the covert bitstream or create detectable artifacts, while lossy transmission, editing, or platform-side normalization can irreversibly damage extraction. As model quality improves, steganalysis systems also improve, producing an arms race in which better generation does not automatically translate into better security.

3.5.7 Critical Unresolved Challenges. Foundational gaps persist across the field: theoretical security guarantees remain unestablished, resilience to advanced attacks is limited, evaluation frameworks lack standardization, responsible development guidelines are absent, non-English performance lags substantially, and real-world deployment testing remains insufficient. Several studies also surface unresolved concerns around reversibility, bias, and misuse potential, indicating that robust engineering alone will not be enough without clearer security and governance models [1, 6, 38, 54].

Quantitative evidence underscores several of these constraints. The PSIC effect creates a trade-off where increasing the embedding rate (e.g., to ~5 bpw) strengthens anti-steganalysis but significantly degrades perceptual text quality [2]. Attack vulnerability is notable; for example, adaptive attacks can reduce bit accuracy to near-random levels (~0.6) [40]. Segmentation ambiguity and entropy variability in communication channels are major impediments that lead to decoding failures and restricted practical capacity [3]. Finally, computational optimizations induce significant overhead, with some reordering algorithms causing up to 69% performance degradation in processing speed [21].

3.6 Contextual Compatibility as a Research Direction

The findings surveyed above indicate a field in transition. Core problems remain unresolved, but the way researchers frame and address those problems has changed. The most visible change in the reviewed literature is the move from predominantly white-box, probability-driven methods toward a broader design space that includes black-box, context-aware, and robustness-oriented approaches. Earlier work depended heavily on local lexical or sampling manipulations, which are precise but difficult to scale and semantically fragile. More recent systems use LLMs to support full-text generation, rewriting, prompting, retrieval, and stylistic control, shifting the problem from isolated token selection toward the joint management of discourse, context, and transmission resilience. Whether this shift also produces more secure systems in practice remains insufficiently demonstrated by current evaluation methods.

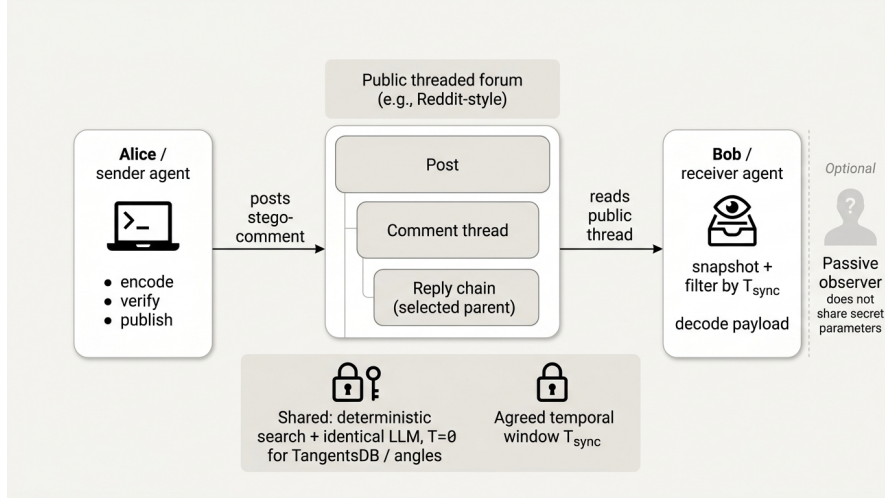


Fig. 2. Covert communication scenario: Alice and Bob use a public threaded channel; synchronization relies on shared operational parameters and a negotiated temporal window.

The PSIC effect (Section 3.5) remains one of the central technical problems in the field. The reviewed studies collectively suggest that **contextual compatibility** can mitigate this conflict, although it does not resolve it. When systems align generated text with the topic, style, and discourse expectations of a target channel—drawing on the external knowledge sources surveyed in Section 3.4—the text can become more plausible to human readers without necessarily becoming more anomalous to statistical detectors. Framed this way, security is not only about distributional proximity to natural text; it also depends on whether the output functions as a plausible communicative act in its intended setting. This is the gap that the context-aware, selection-channel embedding proposed in Section 4 directly targets: rather than treating contextual fit as a secondary quality concern, it is used as the primary embedding domain, with reply targeting and angle selection carrying the payload while the generated comment itself functions as an ordinary, contextually grounded reply.

4 METHODOLOGY

This paper proposes a black-box steganographic technique that embeds information within the semantic architecture of a conversation. The methodology is predicated on a standard covert communication scenario involving Alice (the sender) and Bob (the receiver), communicating over a public social media platform. While this study utilizes Reddit as a primary case study, the underlying principles are platform-agnostic and transferable to any threaded discussion environment. Figure 2 sketches the end-to-end covert scenario and shared synchronization assumptions.

4.1 System Architecture

The core of the proposed framework centers on a multi-layer embedding process that leverages both the *spatial location* of the response and the *thematic subject* of the content as carriers for the secret payload. By mapping binary data to high-level conversational actions—specifically, the selection of a reply target and the selection of an *Angle*—the system carries the payload through a selection channel while preserving perceptual naturalness in the visible text. In the implemented workflow, the receiver reconstructs the same pre-sender context (thread snapshot + deterministic

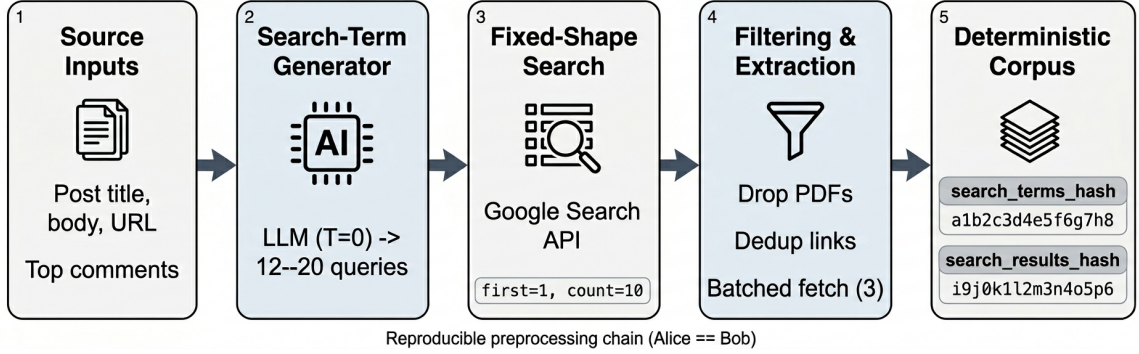


Fig. 3. Deterministic research stage: shared term generation, fixed search parameters, filtering, batched fetching, and integrity hashes for replayable corpora.

research + regenerated angles) and decodes the same selection indices to recover the payload. Figure 6 depicts the two embedding layers and their interaction with synthesis.

4.2 Operational Requirements

To achieve synchronization, both the sender and receiver must maintain access to:

- **Public Communication Channel:** Accounts on the target social media platform.
- **Shared Operational Parameters:** A deterministic search algorithm, post metadata, and predefined search constraints.
- **Deterministic Reconstruction Settings:** A synchronized model/backend configuration for deterministic stages (e.g., temperature=0) so both parties regenerate the same search terms and angle set from the same inputs.
- **Agreed Temporal Window:** A negotiated T_{sync} used to filter thread state so the receiver reconstructs the same parent-comment context seen by the sender.

4.3 Search Determinism & Synchronization Framework

Ensuring that both Alice and Bob operate on the same researched corpus is the cornerstone of reliable decoding. The implemented research stage is a reproducible preprocessing chain rather than an ad hoc web search. A search-term generator first extracts 12–20 unique queries from the post title, body, and URL using the workflow LLM at temperature=0. Each query is then executed through the Google Custom Search backend with a fixed request shape (first=1, count=10), duplicate links and PDF results are removed, and the surviving URLs are fetched in batches of three. Content extraction uses a crawl4ai-based adapter that normalizes URLs and extracts the primary text content. The resulting flat list of fetched page texts is stored as a search-results corpus, and the pipeline records term and corpus integrity hashes so the same corpus can be replayed by downstream stages. Figure 3 summarizes this deterministic research chain.

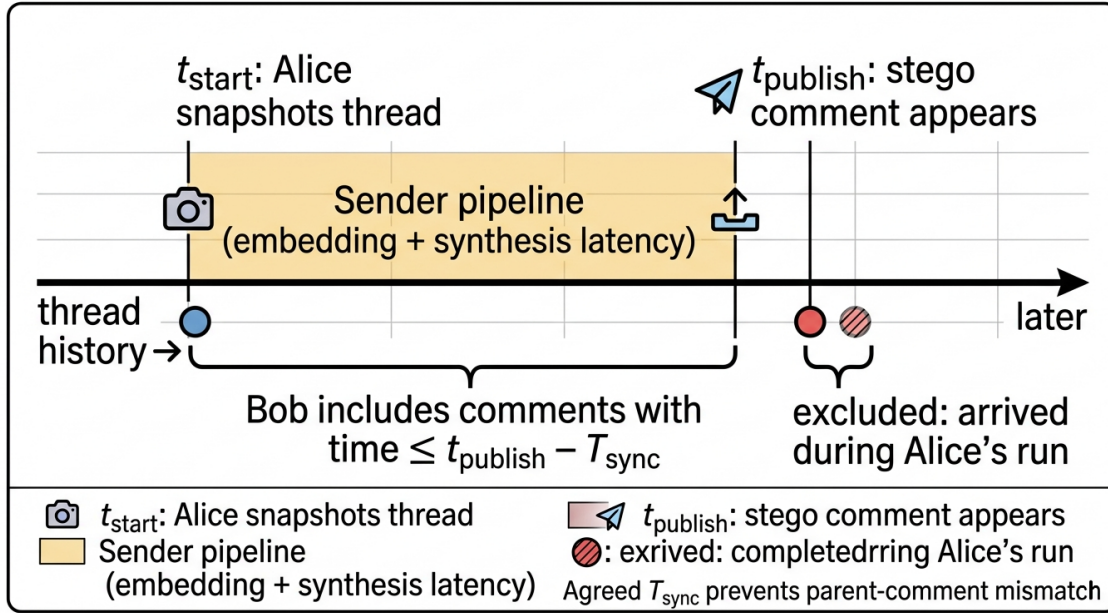


Fig. 4. Temporal synchronization: Bob's reconstructed thread state matches Alice's by agreeing on T_{sync} and filtering comments relative to $t_{publish}$.

4.4 State Synchronization (Temporal Window)

To ensure that both Alice and Bob visualize an identical state of the conversation tree, they must adhere to a negotiated *Temporal Window Duration* (T_{sync}). This synchronization mechanism is operationalized to address the processing latency inherent in the sender's agent:

- (1) **Sender Side:** Alice captures a snapshot of the discussion thread at time t_{start} , performs the multi-layer embedding, and subsequently publishes the generated stego-comment at $t_{publish}$.
- (2) **Receiver Side:** Upon identifying the stego-contribution published at $t_{publish}$, Bob filters the thread's historical state to include only those comments established prior to $t_{publish} - T_{sync}$.

The agreed-upon duration T_{sync} ensures that the "Comment Chain" retrieved by the receiver precisely replicates the context available to the sender at t_{start} , even if new comments were published during the sender's computational cycle. This prevents state divergence and ensures that the decoder can confidently identify the exact parent comment and conversational state used for encoding. Figure 4 shows the temporal window relative to t_{start} and $t_{publish}$.

4.5 Payload Optimization & Compression

Prior to embedding, the raw payload undergoes a transformation layer designed to minimize bitstream length. We implement a *dynamic dictionary-based compression* scheme that uses dynamic programming to minimize the total bit-cost:

- (1) **Dictionary Synthesis:** A session-specific linguistic dictionary is synthesized from the post body, the flattened comment bodies, and the fetched search-results text corpus.

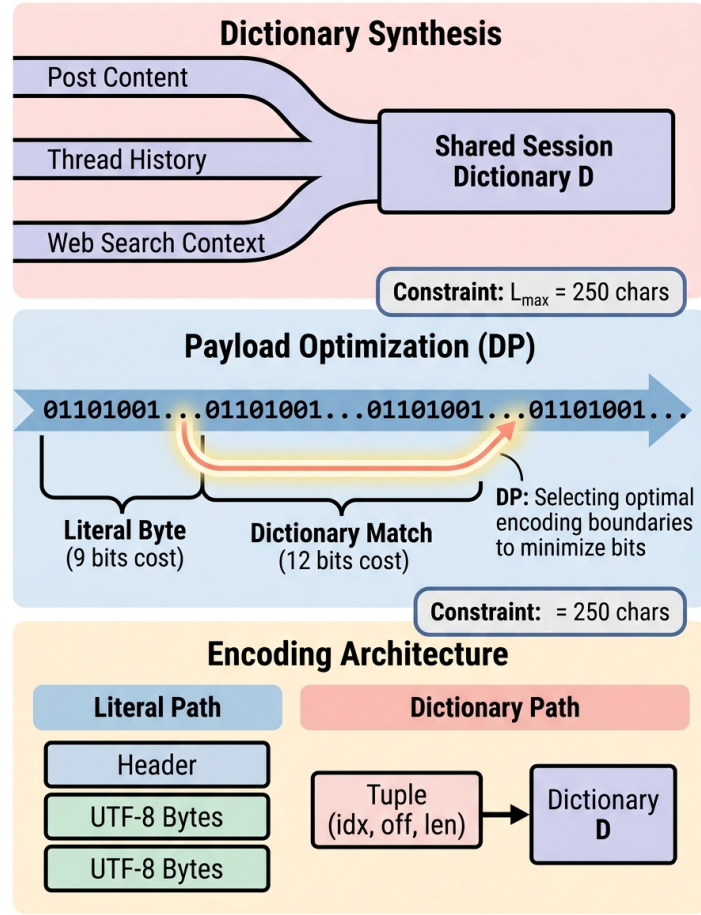


Fig. 5. Payload compression builds a session dictionary from thread and research text, then chooses literal versus dictionary-encoded segments to minimize total bit cost.

- (2) **Optimal Path Selection:** The compression engine scans payload substrings up to 250 characters and evaluates two encoding modes for each segment:
 - *Literal Encoding:* Encodes the raw UTF-8 bitstream with a short length header.
 - *Dictionary Encoding:* Encodes a reference to the dynamic dictionary using a tuple of (*Dictionary Index*, *Offset*, *Length*).
- (3) **Bit-Cost Optimization:** The dynamic program selects the sequence of encoding modes that minimizes the total compressed length. If the dictionary path is not shorter than standard UTF-8, the compressor falls back to the standard representation.

Figure 5 outlines dictionary construction and the literal versus dictionary encoding modes.

4.6 Two-Layer Semantic Embedding Architecture

Figure 6 provides the schematic overview of how the two embedding layers split the payload before synthesis.

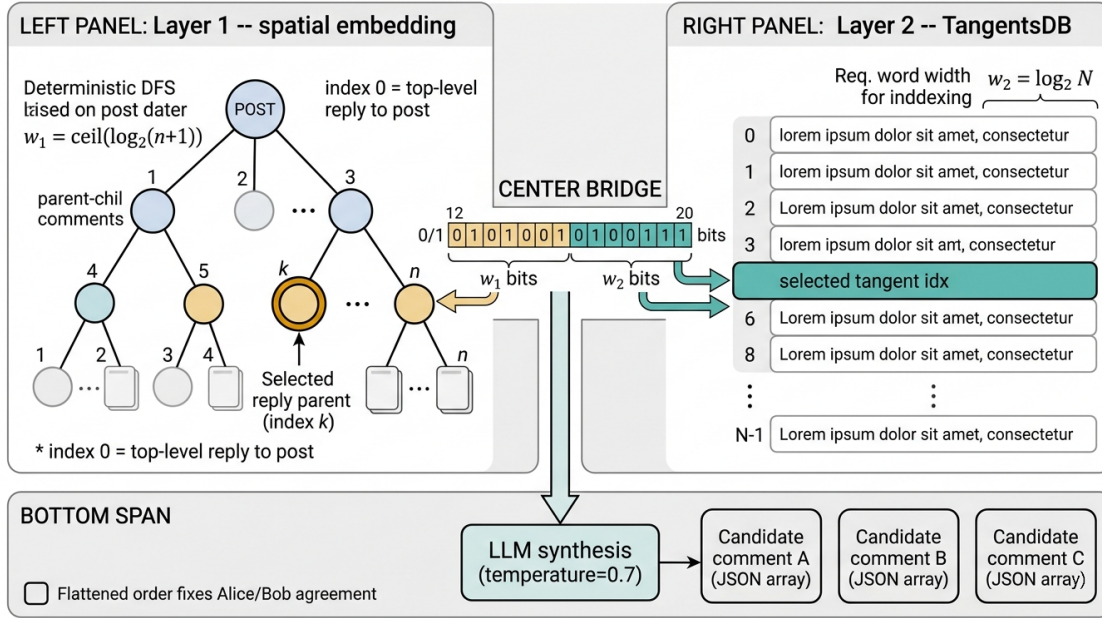


Fig. 6. Two-layer embedding: Layer 1 selects reply placement in a deterministically flattened thread; Layer 2 selects an Angle from a shared, deterministically reconstructed angle set; combined bits drive generative synthesis.

4.6.1 Layer 1: Spatial Embedding (Reply Targeting). The primary embedding layer determines the *architectural placement* of the steganographic contribution.

- (1) The system flattens the nested comment tree with a deterministic depth-first walk.
- (2) The bit-width for this layer is dynamically calculated as $w_1 = \lceil \log_2(n+1) \rceil$, where n is the number of available comments.
- (3) Index 0 corresponds to a top-level reply to the post, while any non-zero index selects a comment whose parent chain is reconstructed through parent-identifier and link-identifier lookups.

4.6.2 Layer 2: Intent-Based Embedding (Angle Selection). The secondary layer determines the *communicative intent*.

- (1) **Angle Set Reconstruction:** Using the deterministic preprocessing chain (Section 4.3) and the filtered thread snapshot, both parties regenerate the same ordered list of candidate Angles, each described by a category label, a source quote, and a tangent field (the thematic direction).
- (2) **Dynamic Bit-Mapping:** The bit-width for this layer is computed from the total number of Angles, and the flattened list is assigned sequential idx values.
- (3) **Angle Selection:** The corresponding payload bits are mapped to a specific Angle index. During synthesis, the selected Angle is paired with the reply context and relevant research excerpts to produce a plausible cover reply.

4.7 Generative Synthesis & Verification

Following the selection of the target context and selected Angle, the sender initiates the cover text synthesis process.

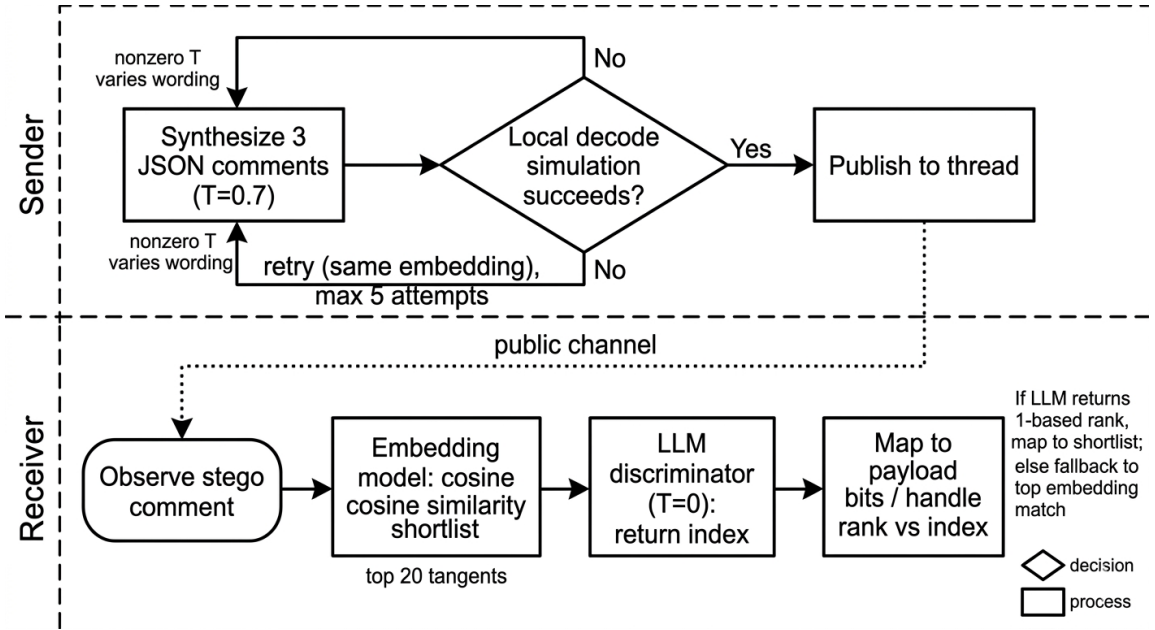


Fig. 7. Sender-side verification retries synthesis until decode succeeds; the receiver recovers Layer 2 via semantic shortlisting and a discriminative identification step.

- **Synthesis:** A configurable generative language model (run at non-zero temperature) produces a small fixed set of short, thread-appropriate reply candidates as a JSON list of non-empty strings. The prompt conditions on post metadata, the selected comment chain, the selected Angle (tangent), and relevant research excerpts.
- **Verification Loop:** Before the stego-text is committed to the public channel, the sender executes a local *Decoding Simulation* to verify that the generated text remains recoverable.
- **Recursive Refinement:** If the simulation fails to recover the intended Angle, the sender re-executes synthesis without changing the embedding parameters. The non-zero temperature ensures output variation; the retry loop is capped by a fixed attempt budget.

Figure 7 summarizes sender-side verification and receiver-side recovery.

4.8 Decoding & Intent Recovery

To recover the hidden intent (Layer 2), the receiver first reconstructs the pre-sender context deterministically, then decodes the Angle selection from the visible reply:

- (1) **Context Reconstruction:** The receiver locates the candidate stego comment, removes it from the thread, applies the temporal cutoff ($t_{publish} - T_{sync}$), and reruns deterministic preprocessing (URL normalization, research, and angle generation) to reproduce the same ordered Angle list.
- (2) **Semantic Filtering:** A semantic similarity model shortlists the top- N Angles whose text descriptions best match the observed stego reply.
- (3) **Discriminative Identification:** A deterministic identification prompt is executed on the shortlisted candidates to return the most likely Angle index:

"Identify the specific Angle from the provided candidates... return the index only."

If the discriminator returns a 1-based rank rather than a raw index, the pipeline maps the rank back to the shortlist; otherwise it falls back to the top semantic match. Combined with Layer 1 (reply targeting), the recovered indices reconstruct the same selection bits and allow recovery of the compressed payload.

5 IMPLEMENTATION

The practical implementation follows the same stage order as the workflow runner: data load, research, angle generation, stego synthesis, and decode. Each stage persists artifacts to disk so later stages can reuse the exact same corpus and hashes, which keeps the sender and receiver aligned across retries.

5.1 Architectural Components

The system is operationalized through a series of deterministic pipeline stages. Core components include:

- **Data load stage:** Fetches unresolved URL content and replaces the post body with the extracted text.
- **Research stage:** Generates search terms from the post title, text, and URL using a deterministic LLM configuration (e.g., `temperature=0`), queries a fixed-shape search backend, removes duplicate links and PDFs, and fetches page text in bounded batches. The stage persists a flat search-results corpus together with reproducibility hashes.
- **Angle generation stage:** Flattens the corpus assembled from the post text, search-results corpus, and comments, then calls the angle runner to produce triplets consisting of a category, a source quote, and a tangent. Long inputs are split into 30k-character segments and batched up to 150k characters. If the backend path fails, the pipeline falls back to a direct LLM prompt that uses the same JSON schema.
- **Stego and decode stages:** Combine the compression layer, comment targeting, semantic shortlist, and verification loop. The encoder generates a small fixed number of candidate comments for the selected Angle, decodes them immediately, and retries synthesis if the selected Angle is not recoverable.

Figure 8 aligns these components with the staged implementation and on-disk artifacts.

5.2 Model Configuration

The implementation uses different model roles for different stages, but the exact model IDs are treated as configurable parameters:

- **Deterministic stages (term/angle generation):** Run at `temperature=0` to ensure reproducible reconstruction from the same inputs.
- **Stego synthesis:** Runs at non-zero temperature to generate diverse candidate replies for a fixed embedding choice.
- **Semantic matching and decode:** Uses a semantic similarity model for shortlist ranking, followed by a deterministic discriminative identification prompt (e.g., `temperature=0`) to select the final Angle index.

6 EVALUATION

This section reports an artifact-based assessment of the workflow, focusing on (i) embedding capacity of the selection channel, (ii) naturalness and distributional divergence of successful stego samples, and (iii) operational reliability under the sender-side verification loop. We center results on a single clean end-to-end run artifact set and explicitly separate generation/decode failures from infrastructure errors.

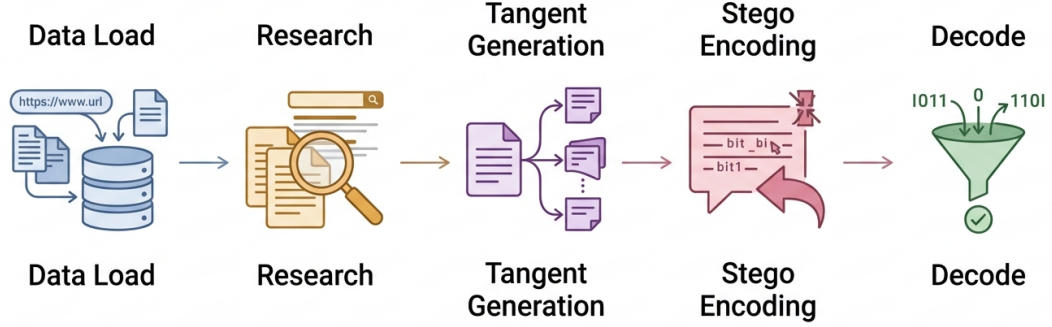


Fig. 8. Implementation stages and persisted artifacts align with the methodology: each stage is deterministic and replayable where required for sender–receiver parity.

6.1 Capacity Analysis

The channel capacity of Layer 2 is fundamentally dependent on the size of the shared, deterministically reconstructed angle set. If the reconstructed set contains N Angles, then Layer 2 contributes $\log_2 N$ bits per message instance (rounded up to an integer bit-width in the implementation). In practice, broader threads and richer research corpora tend to yield larger and more diverse angle sets than narrowly constrained topics, increasing available capacity.

6.2 Perceptual & Statistical Imperceptibility

Imperceptibility is evaluated through the dual lenses of semantic coherence and contextual integration. Unlike traditional steganographic methodologies that risk producing disjointed or statistically anomalous text, our intent-based approach aims to ensure that each stego contribution maintains a clear thematic alignment with the surrounding discourse. In this work we report GPT-2 perplexity and word-unigram KL/JSD as coarse statistical proxies, noting that these metrics do not fully capture long-context semantic quality [49] and do not replace dedicated steganalysis benchmarks.

Table 9 summarizes perplexity (GPT-2) and word-unigram divergence against two within-run reference distributions: the matched primary post and a global comment corpus. Values are computed over the successful, receiver-validated stego samples from a single clean end-to-end run (requested=200, succeeded=183).

Definitions. Perplexity is computed from average per-token log-probability, $2^{-\frac{1}{n} \sum_{i=1}^n \log_2 P(x_i | x_{<i})}$: lower PPL indicates more natural continuations under the model; higher PPL indicates surprising or awkward text. The observed maximum (~ 654) likely reflects a small number of pathological samples that are highly unnatural to GPT-2.

Table 9. Perplexity and distribution divergence for stego comments (GPT-2 PPL; KL/JSD on word unigrams).

Category	Metric	Reference set (this run)	Mean	Range (min / max)
Perplexity	PPL (GPT-2)	183 successful stego strings	≈ 142.3	22.8 – 653.8
Divergence	KL	Matched primary post	≈ 9.24	—
Divergence	KL	Global comment corpus	≈ 6.50	—
Divergence	JSD	Matched primary post	≈ 0.590	—
Divergence	JSD	Global comment corpus	≈ 0.588	—

Kullback–Leibler (KL) divergence compares stego word unigrams P_S to a baseline P_C via $\sum_w p_{\text{stego}}(w) \log \frac{p_{\text{stego}}(w)}{p_{\text{baseline}}(w)}$; lower KL indicates closer statistical match to the baseline. KL being lower against the global corpus than against the matched post suggests stego wording drifts toward generic phrasing rather than thread-specific vocabulary.

Jensen–Shannon divergence (JSD) is the symmetric, bounded counterpart, $D_{\text{JS}}(P_C \parallel P_S) = \frac{1}{2} D_{\text{KL}}(P_C \parallel \frac{P_C + P_S}{2}) + \frac{1}{2} D_{\text{KL}}(P_S \parallel \frac{P_C + P_S}{2})$, often used to upper-bound distinguishability from the true language distribution. Word pairs absent from one side of the comparison are handled with Laplace-style smoothing (e.g., $\alpha = 10^{-6}$) so KL and JSD remain well defined.

6.3 System Robustness & Recovery Rate

The sender-side verification loop improves operational reliability by rejecting candidates that do not decode back to the selected Angle. In a representative clean run (requested=200), the pipeline produced 183 successful, receiver-validated samples and 17 failures after exhausting the retry budget. The primary failure modes were (i) decoding validation failures (10/17) and (ii) malformed synthesis output that did not match the required JSON schema (7/17). These failures highlight that reliability is primarily limited by the ability to synthesize a decodable witness under stochastic generation, rather than by payload recovery after validation.

6.4 Resistance to Automated Steganalysis

The design objective of the framework is to reduce semantic and pragmatic anomalies by conditioning synthesis on the full thread context and by encoding information in conversational decisions (reply targeting and Angle selection) rather than local token artifacts. We therefore expect detection to require thread-level pragmatic reasoning rather than relying solely on token-level distribution cues [4]. A dedicated steganalysis benchmark evaluation is left to future work; here we limit claims to the measured distributional and operational metrics reported above.

7 CONCLUSION

This research introduces a context-aware semantic steganography framework for threaded discussions that redefines the embedding domain as a *selection channel*. By transitioning from token-level manipulation to encoding payload bits into reply targeting and Angle selection, the method aims to preserve contextual naturalness in the visible text while retaining black-box compatibility with LLM APIs.

The implementation includes a closed-loop verification mechanism that simulates decoding prior to publication and retries synthesis under stochastic generation. Our artifact-based evaluation reports naturalness and unigram divergence metrics on successful, receiver-validated samples, and characterizes the dominant failure modes when retries are exhausted. Future work will focus on improving end-to-end success rates on fresh real threads, expanding clean sample sizes, and performing dedicated steganalysis evaluations under explicit adversarial models.

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